

The content extracted is as follows:

Cost effective filtering of unreliable proximity detection results based on BLE RSSI and IMU readings using smartphones

This requires appropriate analysis of sensor signals and radio signal strength (measured as RSSI) to determine the appropriate activities and to indicate the correct moments of interaction and filter the remaining ones.

Our approach to proximity detection is based not only on data from the external world (RSSI from static points anchors) but also on information about the internal state of the object under study (expressed by the readings from the Inertial Measurement Unit (IMU) of a smartphone).

We use smartphones (which already have all of the necessary hardware functionalities and only a simple application is needed) to gather data from both static and dynamic objects, which makes our approach cheaper and easier to deploy than the one based on dedicated iBeacons and wearable sensors.

In¹⁸, a complex hybrid localization algorithm is described, consisting of pose and velocity estimation based on activity detection and accelerometer readings and using RSSI signal strength.

Generally, systems created for the purpose of localization and positioning are based on computationally-demanding methods and could not be deployed in environments, in which data from numerous nodes is analyzed (e.g. Generalized Regression Neural Network and Kalman filter in²¹ or Denoising Autoencoders in⁹).

The paper²⁶ describes a method for recognizing basic activities, based on smartphone inertial sensors accelerometer and gyroscope.

The alternative to these datasets can be some of the PACT datasets³², which consist of Bluetooth RSSI data.

Unfortunately, the sensory data is gathered independently of Bluetooth frames (therefore it is impossible to combine the data regarding RSSI and sensor readings) and only static activities were considered. Due to the fact that none of the datasets in the literature meets all of the requirements of the methods proposed in the article, dedicated dataset had to be used in the experiments.

As an answer to the problem formulated in Problem and aim definition, we propose a system based on Bluetooth technology and IMU readings from smartphones for detecting proximity to a static point (some point of interest) and evaluating the quality of this prediction (based on determining the degree of dynamics of a moving object (e.g., a person) based on introduced metrics obtained based on predictions of a lightweight and efficient machine learning model for activity recognition and Bluetooth RSSI readings from the surrounding static points.

They also found application (and they were proven successful) in works considering different types of human activity recognition (e.g. classification of alpine skiing styles based on IMU data and accelerometer-based human activity recognition in works³ or⁴).

Collecting data in a way that depends on the reception of Bluetooth frames, we get RSSI values and sensor readings related to a single timestamp.

Data processing. To estimate the proximity and build the metrics that can be used to assess the reliability of proximity detection, we have to first process raw RSSI and sensor readings to obtain the data necessary to run our algorithms.

We calculate the average value of the label over n frames the final table will be approximately n times smaller, because RSSI reading for each sender have to be obtained for particular time moment this way we obtain more stable values of position estimation over time. The activity_metric can be defined as an average prediction of an activity prediction model in a time window.

All of these metrics can be used to filter out the moving points in the proximity detection task and to assess the reliability of the detection (especially proximity between a dynamic and static point while the dynamic object is moving towards some different location) and are easy to interpret higher values of activity_metric and rssi_metric suggest that movement of a person can be involved, whereas the higher values of prediction_reliability_metric values suggest that it is more probable that the examined person was moving between points during the time taken into consideration.

Alternatively, the values can be chosen by the operator (especially the second value specifying the proportion in which we consider metrics based on RSSI and on activity predictions (see Fig.

6a,b, and 7 we presented the filtering results for the three examined metrics (activity_metric, rssi_metric and reliability_metric respectively).

Time windows, for which the value of a particular metric are higher than the threshold (for RSSI and activity-based metrics) or lower than the threshold (reliability_metric) are treated as unreliable and should be filtered (red crosses in the figures).

The results obtained for the activity_metric (based on IMU readings, see Fig.

If we use a metric based on activity and RSSI (see Fig.

Due to the use of simple models for activity detection and simple methods for determining the value of RSSI-based metric, the solution is easy to implement, robust, memory and resource efficient and highly scalable.

Values of importance coefficients (and in Eq. 8) can be determined via simple optimization process, in which we aim to find such a value of one of the coefficients that allows us to obtain the maximum value of correlation between the reliability_metric and the labels representing the occurrence of a movement (we labelled our data manually based on the course of the experiment, see Fig.

It is also possible to use only one of the base metrics tested (activity-based or RSSI-based) or to take only one of these metrics into account when creating a joint metric (to take only the activity prediction into account, set the importance to 1, and to take only the RSSI-based metric into account, set it to 0).

When a metric based only on RSSI is used for movement detection then there is no need to use IMU data and Machine Learning model and an even more computationally and memory efficient solution can be achieved (a mild trade-off between accuracy and efficiency, as the results for RSSI metric are also acceptable).

Therefore, our motion filtering solution can be used to filter out the results for distance estimation or localization (RSSI data can be aggregated only over a time window when the object remains in the same location no movement is observed).

It is the reason why we assign larger weights to closer static points for determining the RSSI-based activity metric.